

Kaidi Zha

UNDERGRADUATE RESEARCHER · AI & MACHINE LEARNING

Department of Physics, Tsinghua University, Beijing, China

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“Building intelligent systems that see, learn, and act.”

Research Interests

Multimodal Large Models, Efficient Deep Learning

Model Architecture Design, Data-Centric AI, Reinforcement Learning

Dedicated to building more efficient models through **novel model architecture design** and **data-centric strategies**, with a focus on improving **training efficiency**, **inference throughput**, and **data utilization**.

Education

Tsinghua University

Beijing, China

B.S. IN MATHEMATICS AND PHYSICS

Sept. 2023 - Present (Expected 2027)

Overall GPA: 3.5/4.0.

RELEVANT COURSEWORK

Mathematics & Physics

Applied Stochastic Processes (A-, 4.0), Probability and Statistics (A, 4.0), Numerical Analysis (A, 4.0), Mathematical Physics Equations (B+, 3.6), Complex Analysis (B+, 3.6), Quantum Mechanics I (B+, 3.6)

Computer Science

Data Structures & Algorithms (A-, 4.0); in progress: Pattern Recognition & Machine Learning, Statistical Machine Learning, Reinforcement Learning & Controls, Fundamentals of Computer Graphics

Research Experience

Model Compression: DiT Post-Training Quantization

Supervised by Prof. Yuan Meng,
Dept. of CS & Tech, Tsinghua
University

RESEARCH ASSISTANT

Jun. 2025 - Present

- **Contributions:** Proposed a layer-wise mixed-precision quantization scheme; refactored the entire codebase for modular evaluation; conducted all experiments on ImageNet-1K.

Supervised by Borong Lin & Yihui Li,
School of Architecture, Tsinghua
University

Data for Buildings: Large-Scale Graph Datasets

RESEARCH ASSISTANT

Apr. 2026 - Present

- **Contributions:** Validated dataset quality by running experiments under diverse model architectures and hyperparameter settings.

Publications

ArchEGraph: A Large-Scale Graph Dataset for Geometry-Topology-Physics Aligned Building Energy Modeling

under review at D & B Track

YIHUI LI, YIHUI CHEN, **KAI DI ZHA**, XIAOYUE YAN, ZHEXUAN YU, SHIQI DAI, JUN XIAO, JUN YIN, RAMON ELIAS WEBER, BORONG LIN

2026

Skills

Programming

Python, C/C++, MATLAB, 数学

Deep Learning

PyTorch, Jax

Tools

Git, Linux, Docker

Languages

Chinese (Native), English (CET-4: 596, CET-6: 534)